Status	Finished
Started	Thursday, 15 October 2026, 10:16 PM
Completed	Thursday, 15 October 2026, 10:21 PM
Duration	5 mins 52 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of all keys
- b. A sorting algorithm is stable if it preserves the order of duplicate keys
- c. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting makes it harder to search for items
- b. Sorting decreases the efficiency of selection operations
- c. Sorting helps in finding duplicates quickly
- d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **3**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes code execution slower
- c. Makes data harder to manage
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Binary Search
- b. Bubble Sort
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is sorted independently
- b. The parts are merged without sorting
- c. Each part is searched for a specific element
- d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **6**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. order()
- c. sort()
- d. arrange()

The correct answer is: sort()

Question **7**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Heap sort
- b. Insertion sort
- c. Quick sort
- d. Bubble sort

The correct answer is:
Insertion sort

Question **8**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Bubble Sort
- b. Quick Sort
- c. Merge Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Searching
- c. Duplicates
- d. Selection

The correct answer is: All of the above

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Selection sort
- b. Heap sort
- c. Quick sort
- d. Merge sort

The correct answer is:
Merge sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It repeatedly finds the minimum element and moves it to the sorted part of the list
- c. It compares adjacent elements and swaps them if necessary
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **12**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 40,80,90,150,200,300,400,470,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **13**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It complicates the analysis process
- b. It makes the analysis slower
- c. It allows for quicker identification of duplicates
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It avoids the need for recursion
- c. It simplifies the sorting process by using only one pass
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **15**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Sorting the entire list in one pass
- c. Bubbling up the smallest element
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position